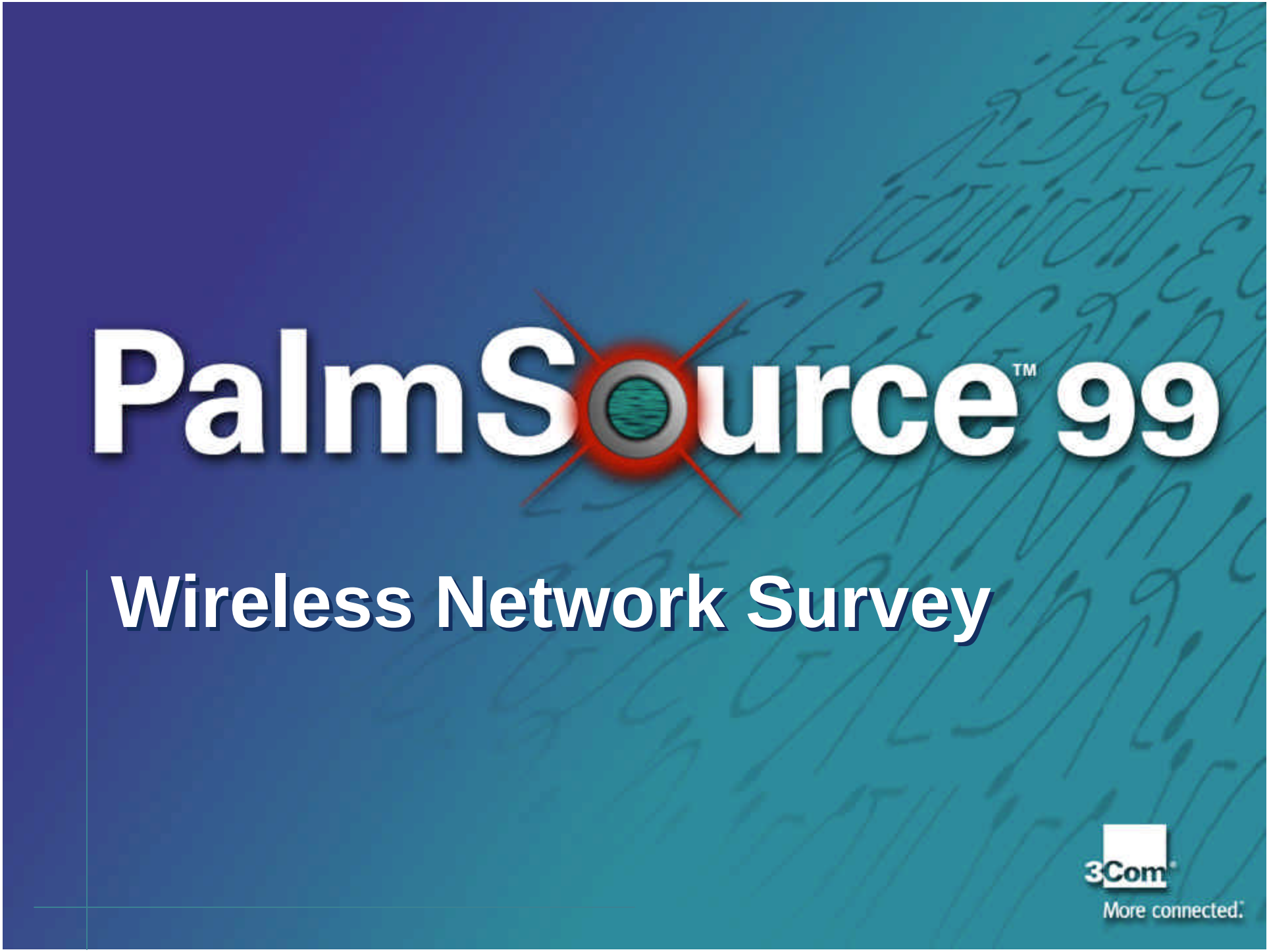

PalmSource™ 99
Enterprise Opportunities for Wireless and the Internet
Steve Butsch
Manager, Systems Engineering
3Com
More connected.

Agenda

PalmSource 99

- Wireless network survey
- Palm™ device as wireless terminal
- Corporate LAN to Wireless WAN Connectivity
- PalmVII™ device and Palm.Net™ service
- Example Implementations
- Q&A


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Wireless Network Survey
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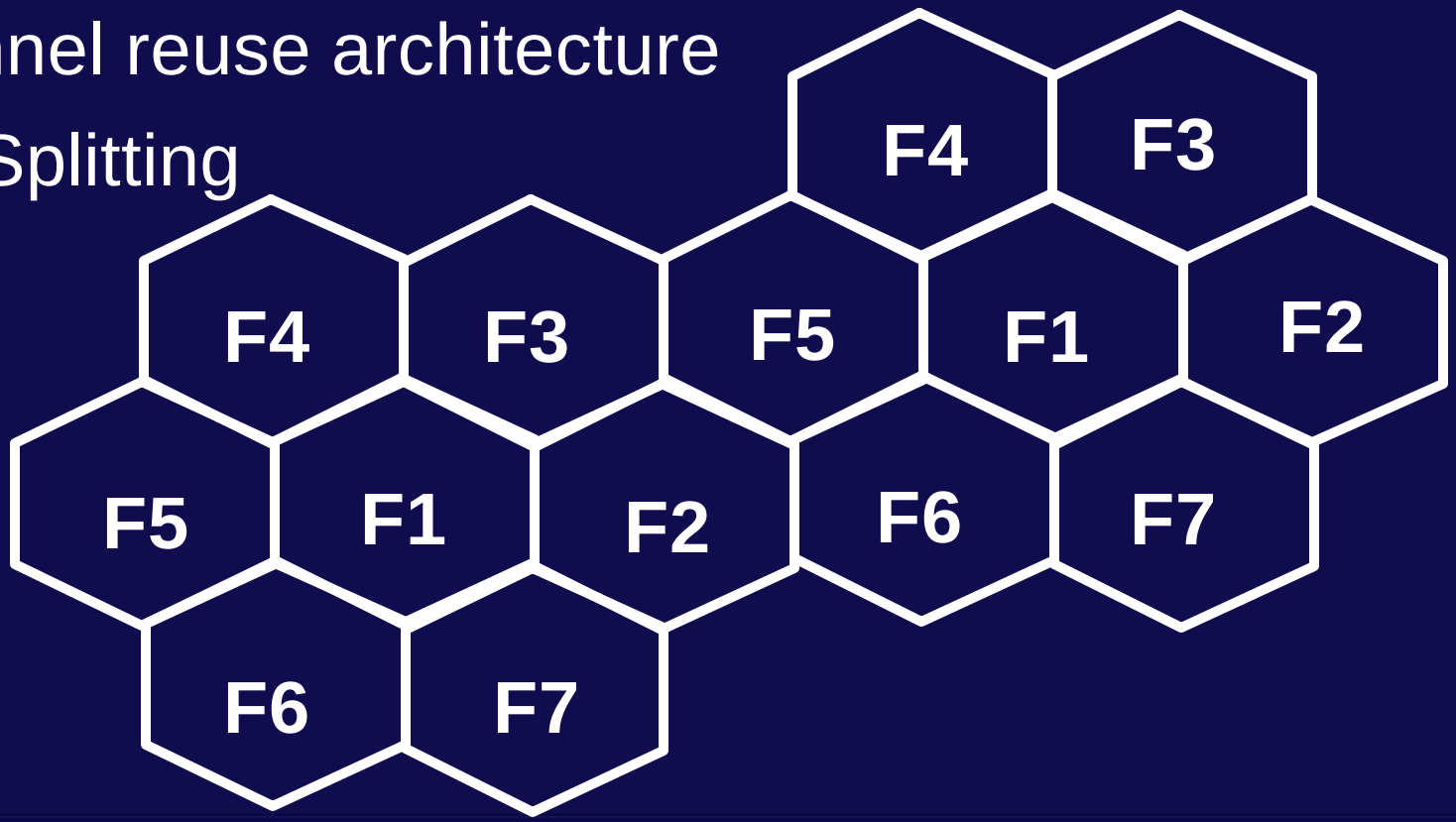
Mobitex

- BellSouth Wireless Data
- Launched in 1992
- Nationwide network
- Mobitex Technology
  - Cell based
  - Packet switched, (MPAK, not IP)
  - 800–900 MHz, 8 KBPS
- Coverage in the top 266 MSAs
  - Translates into top 7800 cities and towns
  - 93% of the U.S. business population

Mobitex: High Capacity Design

PalmSource 99

- 10 to 30 channels throughout the U.S.
- 400 to 600 users per channel
- Channel reuse architecture
- Cell Splitting

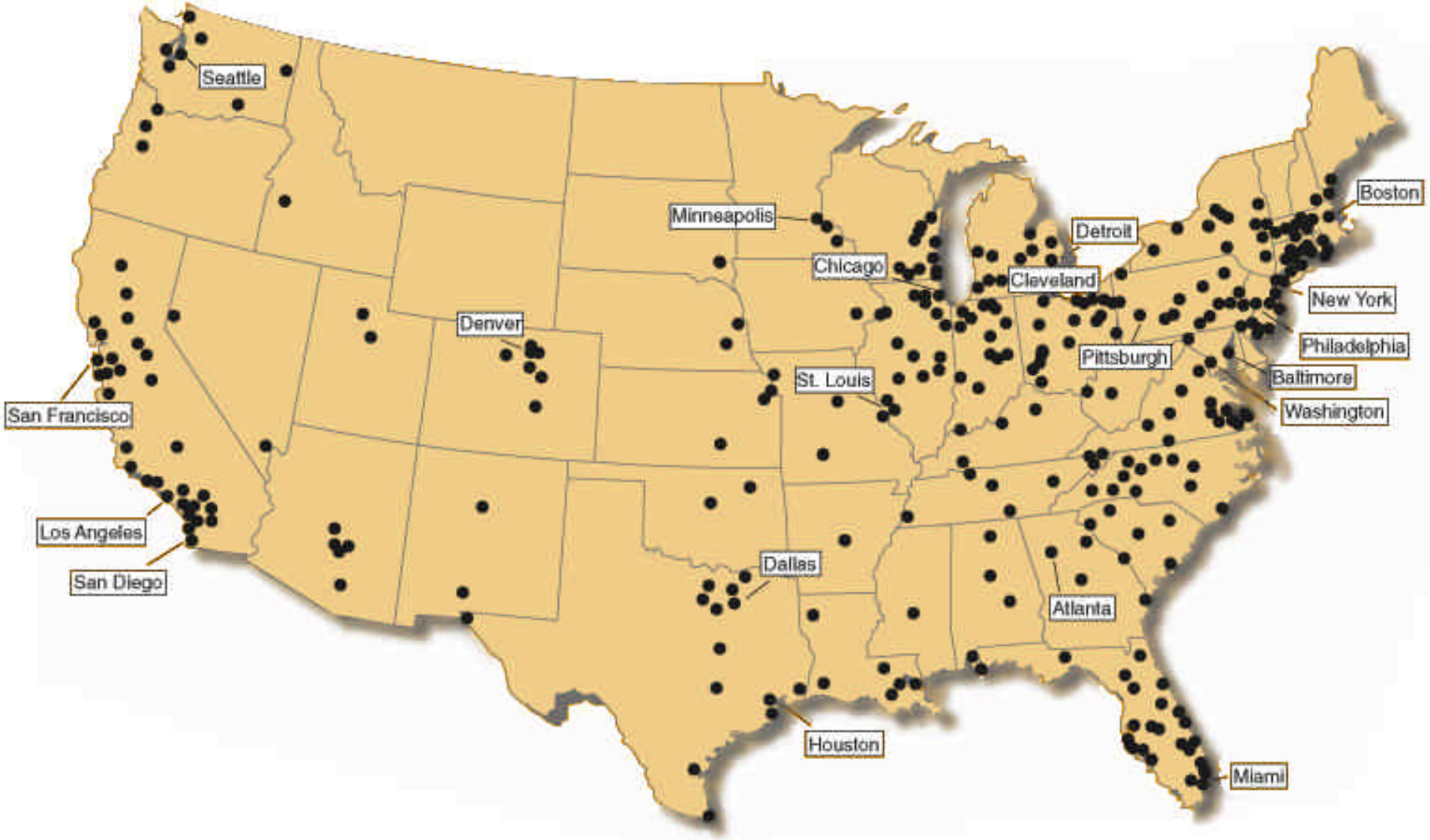


Advantages of Wireless Packet Data

- Better response time for short messaging
- Reliable
- Virtually connected
- Tolerant of coverage gaps
- Spectrum efficient

BSWD Coverage

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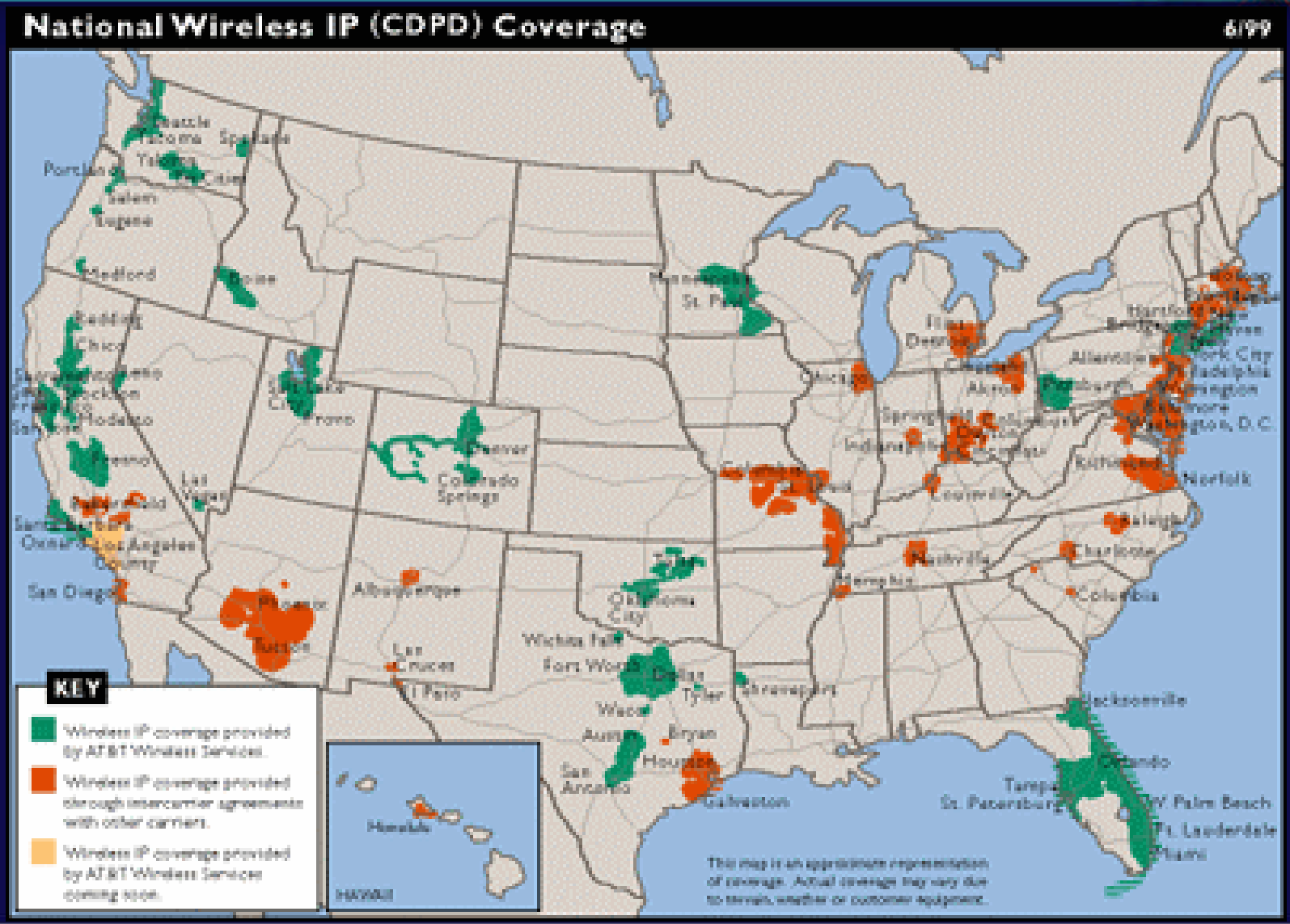


CDPD: Cellular Digital Packet Data

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- Offered by AT&T, Bell Atlantic, GTE Wireless, Ameritech, Southwestern Bell, US Cellular and others
- Launched in 1994
- Regionally based w/ roaming agreements
  - Coverage in top 70 MSAs
- CDPD Technology
  - Overlay on existing analog cellular (AMPS)
  - Packet switched
  - 800–900 MHz, 19.2 KBPS
  - Error correction and encryption
  - IP based

AT&T Nationwide CDPD Coverage



ReFLEX™—Two Way Paging

- Skytel SkyWriter Services
  - Other players: Airtouch paging, Pagenet, Pagemart, Metrocall
- Based on Narrowband PCS
  - Licensed 50 KHz blocks in 900 MHz spectrum
- Motorola FLEX™ protocol family
  - FLEX™
  - ReFLEX™ (ReFlex™ 25, ReFlex™ 50)
  - InFLEX™
- Asymmetric transmit/receive channels
  - Outbound 1.6 to 6.4 KBPS
  - Inbound 0.8 to 9.6 KBPS

Ricochet

PalmSource 99

- Metricom Inc.—network launched 1995
- Wide-area, spread-spectrum network
  - Unlicensed frequency band (902–928 MHz)
  - Microcell Radio infrastructure
  - Packet switched, IP based
  - Modem speed ~28.8 KBPS
- Coverage limited
  - San Francisco Bay area, Seattle, Washington DC
  - Small number of airports and corporate campuses
- Flat rate pricing

Cellular / PCS

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Technology	Classification	Frequency Band
AMPS	Analog Cellular	800 MHz
TDMA (IS-136)	Digital Cellular or PCS	800 MHz, 1900 MHz
CDMA (IS-95)	Digital Cellular or PCS	800 MHz, 1900 MHz
GSM 1900	PCS	1900 MHz

TDMA—
Time Division Multiple Access

PalmSource 99

- TDMS IS-136, operates at 800 MHz, 1900 MHz
- Also known as D-AMPS
- Can co-exist with analog cellular channels
  - Leverage off of analog infrastructure
- Major US Carriers
  - AT&T Wireless, BellSouth and Southwestern Bell

CDMA—
Code Division Multiple Access

PalmSource 99

- Spread Spectrum technology developed by QUALCOMM
- 800 MHz, 1900 MHz bands
- US Carriers
  - Airtouch
  - Bell Atlantic/Nynex
  - GTE
  - Primeco
    - Consortium of Airtouch, Bell Atlantic/Nynex and US West
  - Sprint PCS
    - Consortium of Sprint, Comcast and TCI

GSM—Global System for Mobile Communications

PalmSource 99

- TDMA based
- SMS—Short Messaging Service
  - Allows 160 byte messages
- User specific SIM card allows US/European roaming
- US Carriers
  - Omnipoint, Pacific Bell, BellSouth, Sprint Spectrum, Microcell, Western Wireless, Powertel and Aerial
- Over 215 operators in 100 countries

Cellular / PCS—Future

- GRPS
  - Packet upgrade to GSM, IP based, >100 kbps
- CDMA2000
  - Upgrade to CDMA, higher data rate
- Edge
  - Packet upgrade to TDMA (IS-136), > 300 kbps
- IMT-2000 (3G)
  - Convergence of TDMA, CDMA and GSM
  - 144 kbps, 384 kbps and 2 Mbps

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Palm™ Device as Wireless Terminal

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PalmVII™ Device

PalmSource 99

- BSWD Network
- A Palm Computing® platform device
- Instant access to Internet data
- Wireless Internet messaging



Minstrel Wireless IP Modem

- Novatel Wireless Inc.
- Compatible with all versions of the PalmIII™ and PalmV™ device form factor
- Bundled with POP3 client and text-based web browser
- CDPD wireless technology
- Compatible with most ISPs



Unilinear Parachute

- PC Card adapter for Palm
- Cellular-ready modems
- High speed (33.6/56K) modems
- Paging PC-cards
- Wireless modems (RAM, CDPD, and more)
- GPS cards



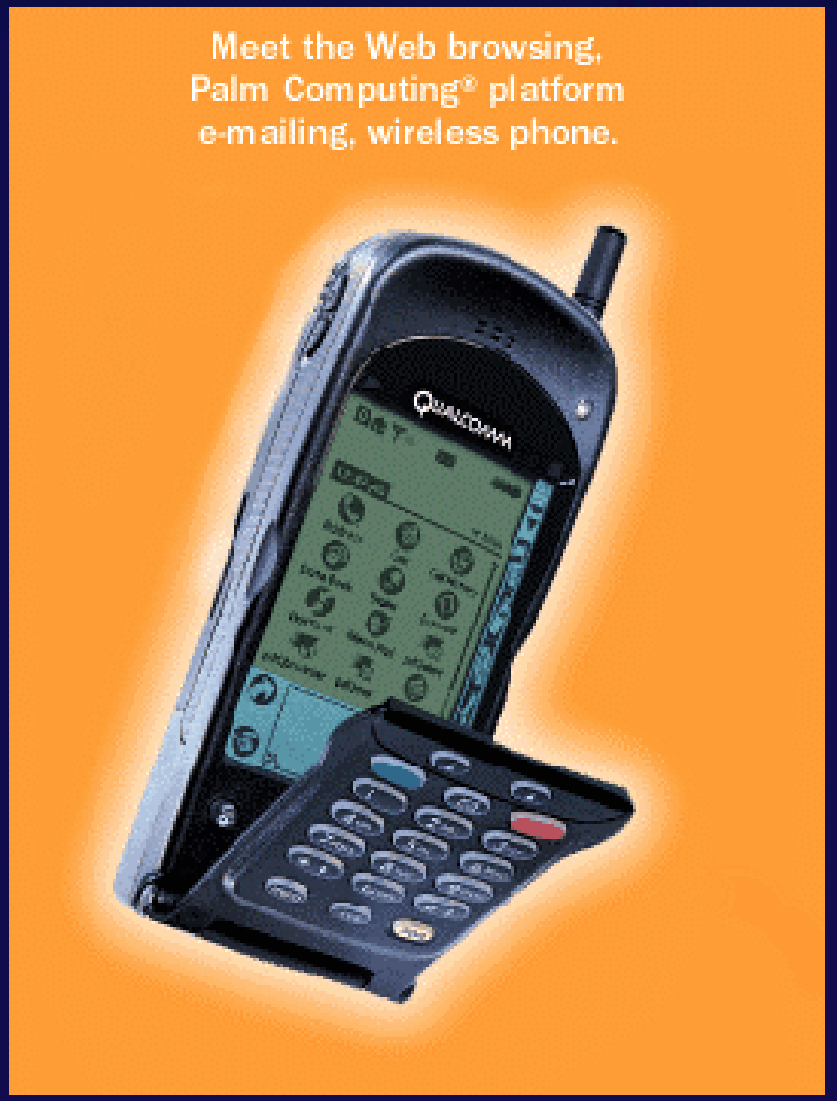
JP Systems BeamLink

- Supports SkyTel's SkyWriter or MCI Interactive Service
- Two-way and one-way messaging
- Send/Receive E-mail
- Group pages and E-mail
- Uses Palm address book



Qualcomm pdQ Phone

- CDMA technology
  - Up to 14.4 Kbps
- Palm OS compatible
  - 2 MB memory
  - 160x240 display
  - Code Warrior application development
- Web, e-mail and paging
  - pdQ 800
  - pdQ 1900



Option GSM Adapter

PalmSource 99

- GSM phone adapter for Palm™ devices
- Supports over 40 mobile phones including Ericsson, Nokia, AT&T and Sony
- Supports remote HotSync® technology, all IP products (Smartcode, ActualSoft, AvantGo, Riverbed etc.)
- SMS support with Handphone application



OPTION
INTERNATIONAL
Mobile Data Communications

Ericsson SH888

- GSM phone
- Built in IR port
- Enables all Palm IP-based applications
  - E-mail
  - Web
  - Network HotSync



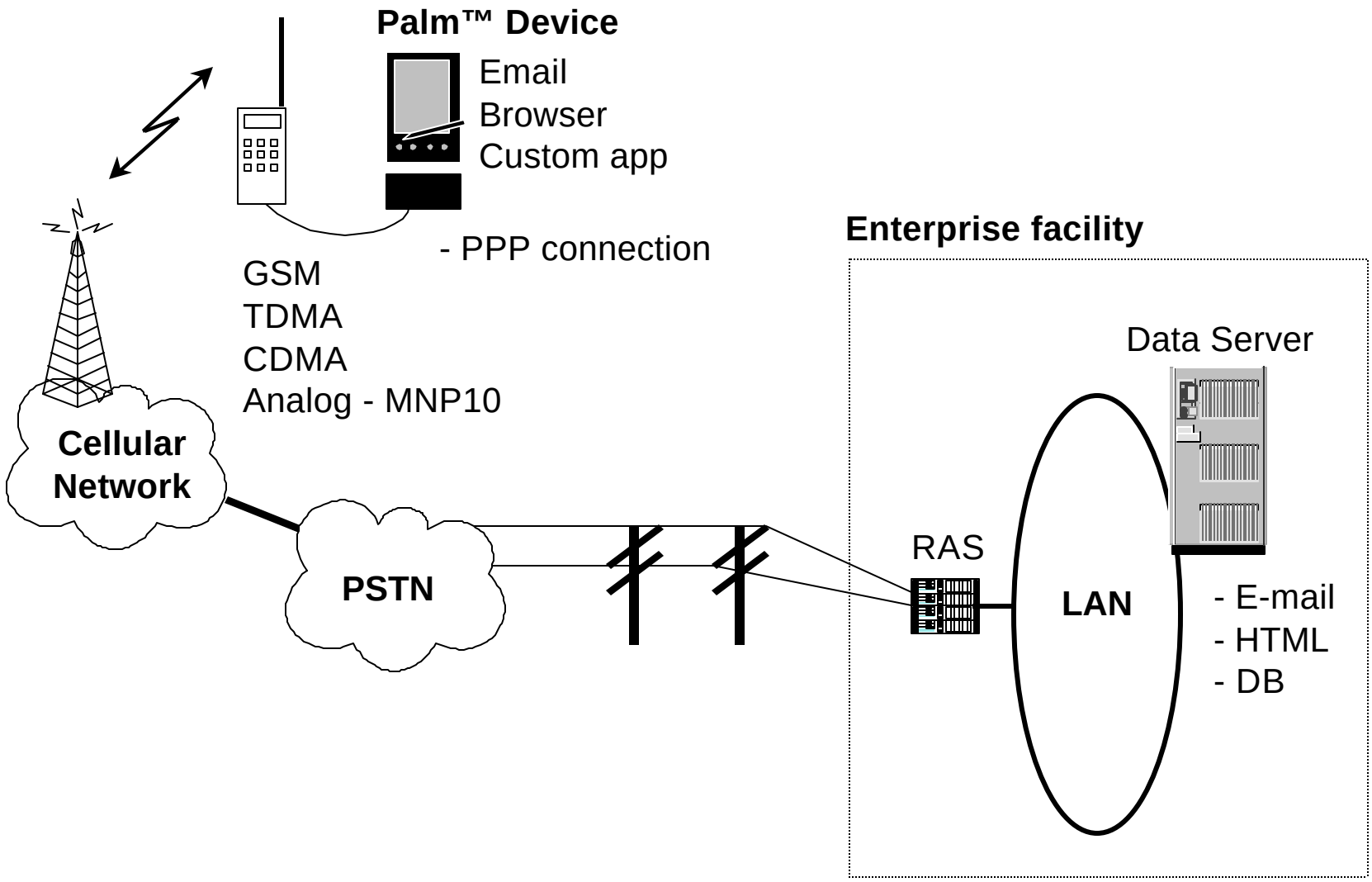
Palm to Ricochet

- Ricochet SE Modem and cable
- Enables all Palm™ IP-based applications
  - E-mail
  - Web
  - Network HotSync

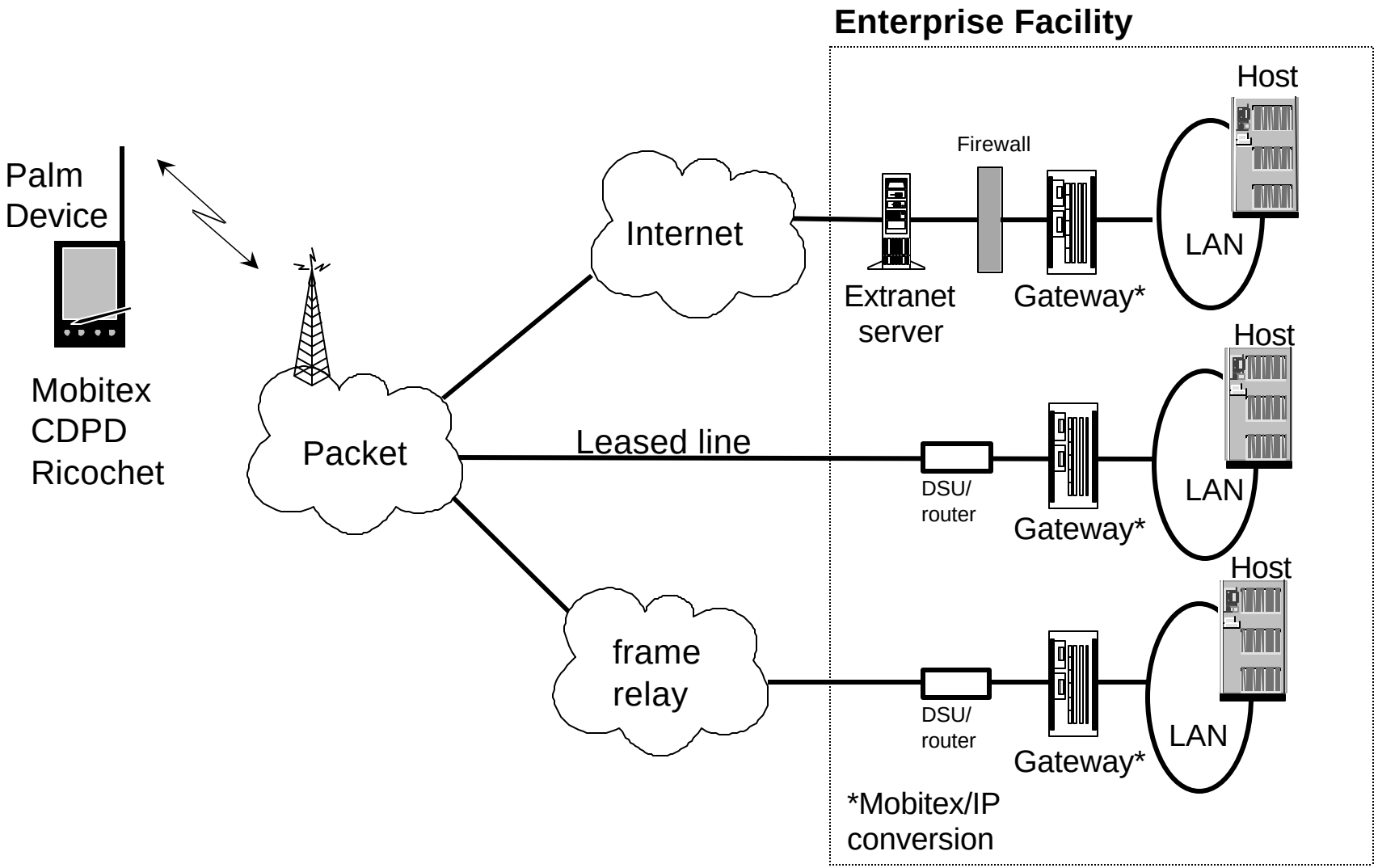



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Corporate LAN to Wireless WAN Connectivity
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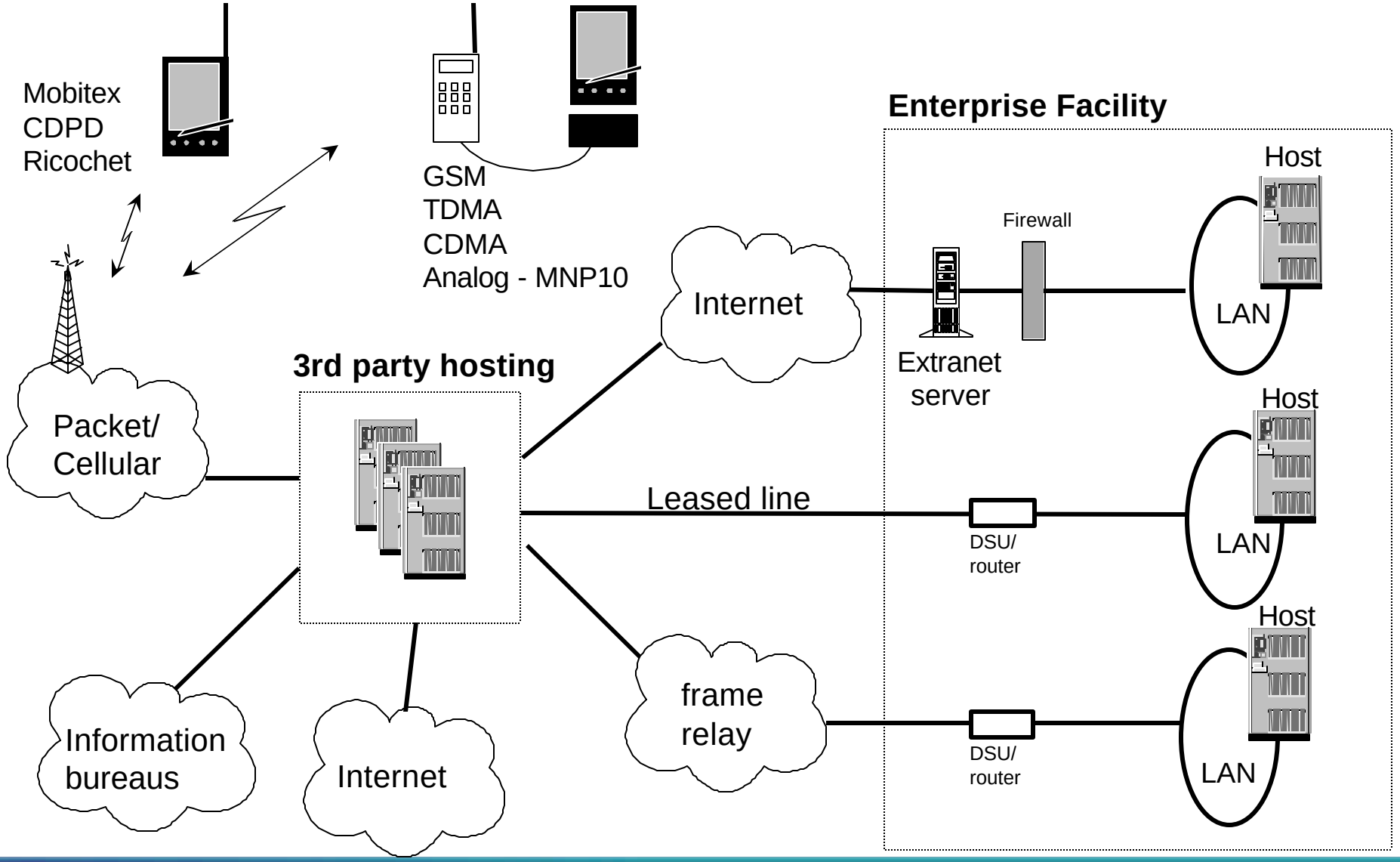
Connecting Corporate LAN to Wireless—Cellular



Connecting Corporate LAN to Wireless—Packet



Connecting Corporate LAN to Wireless—3rd Party Hosting



3rd Party Hosting Services

PalmSource 99

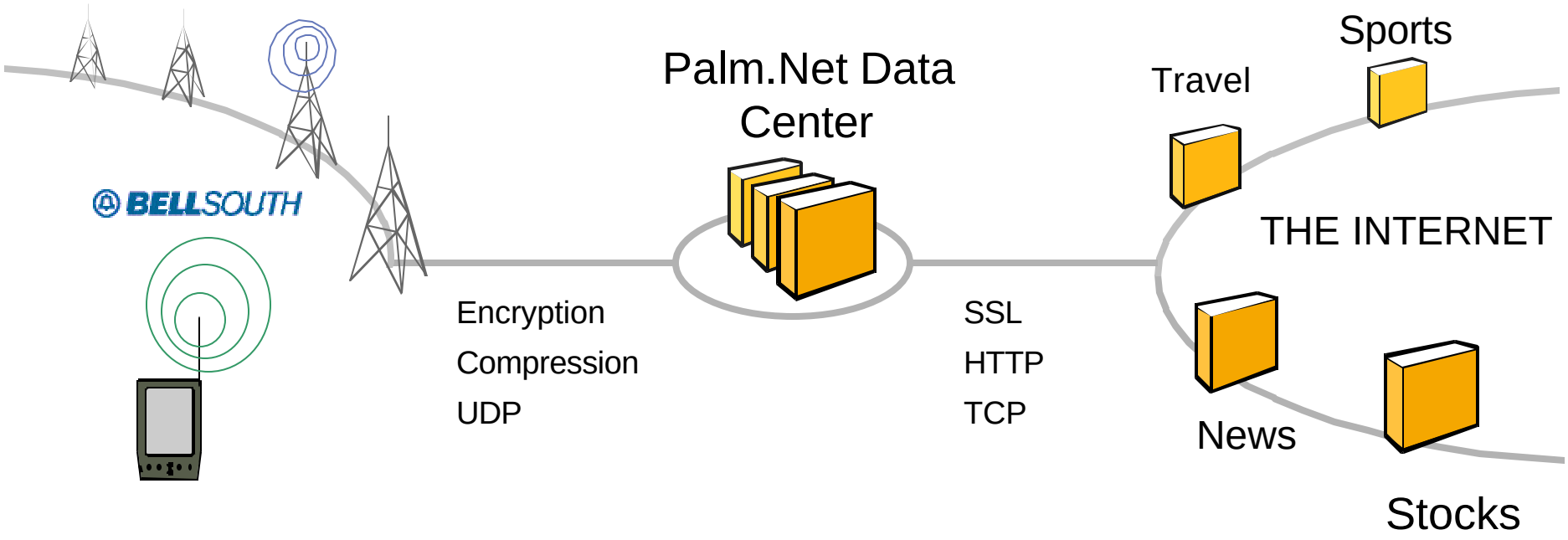
Company	Services	Networks Supported
Aether Systems	Enterprise hosting, info services	BSWD, CDPD, PCS, ARDIS, PSTN
Go America	Enterprise hosting, E-mail, Web	BSWD, CDPD
OpenSky	E-mail, Web, info services	CDPD
Palm.Net	PalmVII device support, Web Clipping, messaging	BSWD

PalmSource™ 99

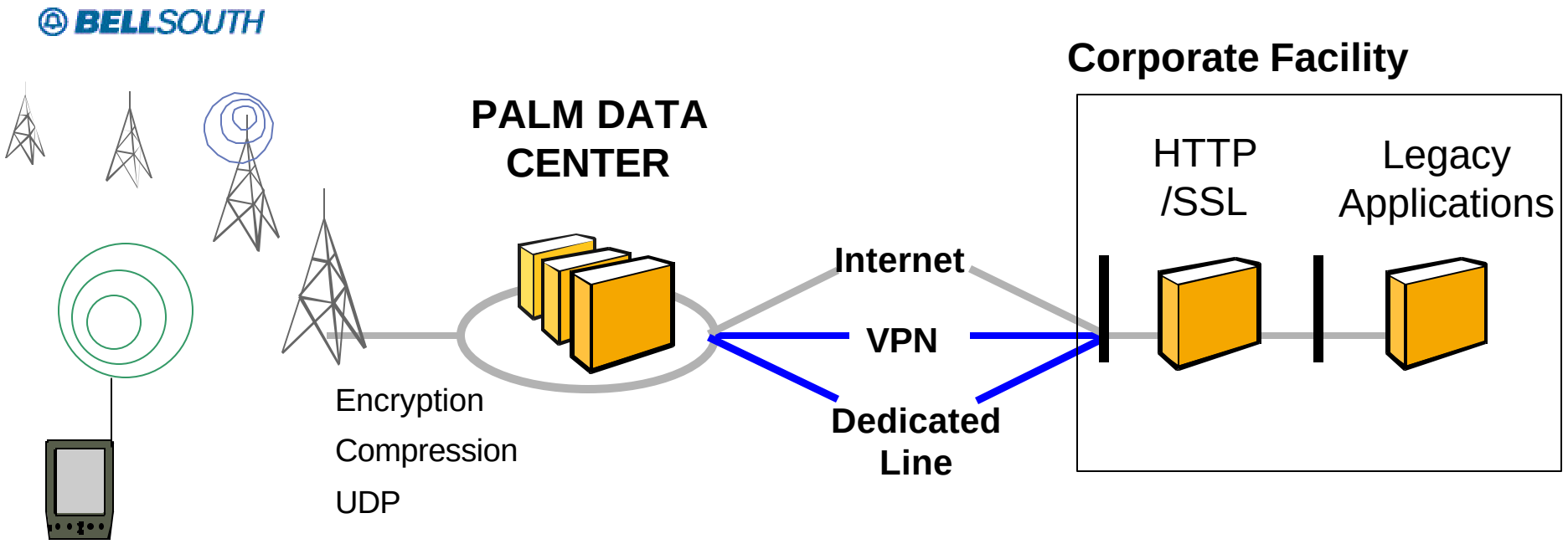
PalmVII and Palm.net

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Palm.Net—Internet Access

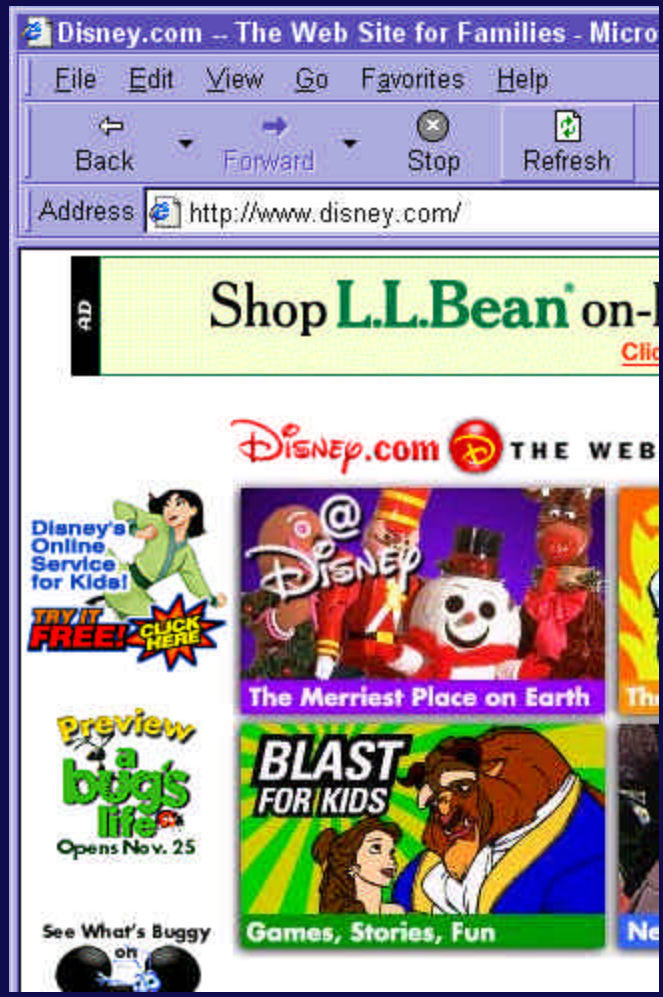


Palm.Net—Enterprise Data Access



Why Browsing Doesn't Work

- Poor performance—relatively low bandwidth and high latency
- Cost in both time and money
- Who wants to "click through?"
- 640x480 format
- Huge battery requirements



Web Clipping—Not Web Browsing

PalmSource 99



Launcher



Query Application

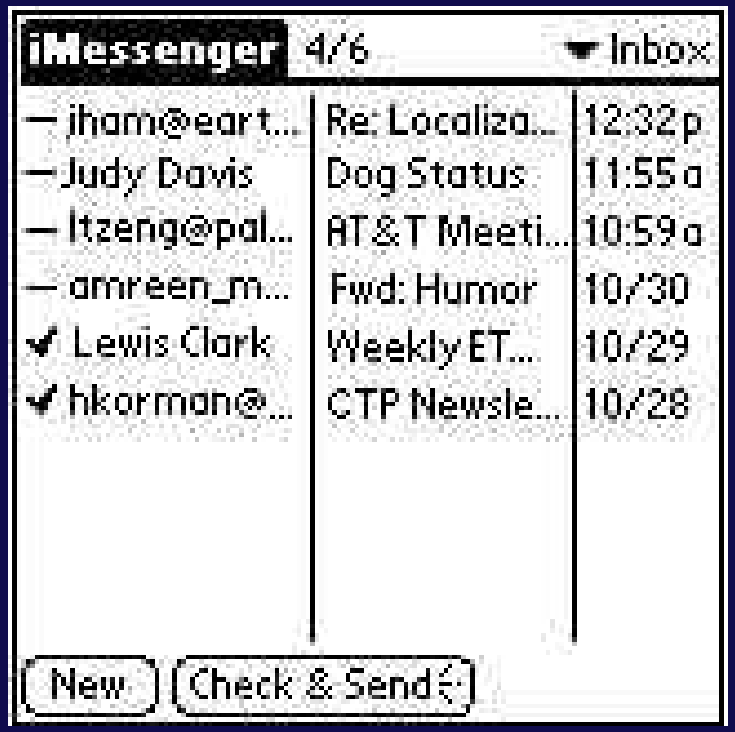


Clipping

Palm.Net—
Wireless Internet Messaging

PalmSource 99

- Anytime message access
- Anytime outgoing communication
- Cell phone model



Palm.Net Security

- Activation methodology required industrial strength
- Partnership with Certicom
- Elliptic Curve Cryptography
- Public/Private Key mechanism
- Equivalent strength of SSL



Developing PalmVII Applications

- Use standard authoring tools
- In some cases, no changes
- In most cases, some changes:
  - New HTML tags
  - Use existing Palm OS extensions
- Palm Friendly—optimal solution
  - Style/design recommendations to optimize viewing of content



PalmSource™ 99

Example Implementations

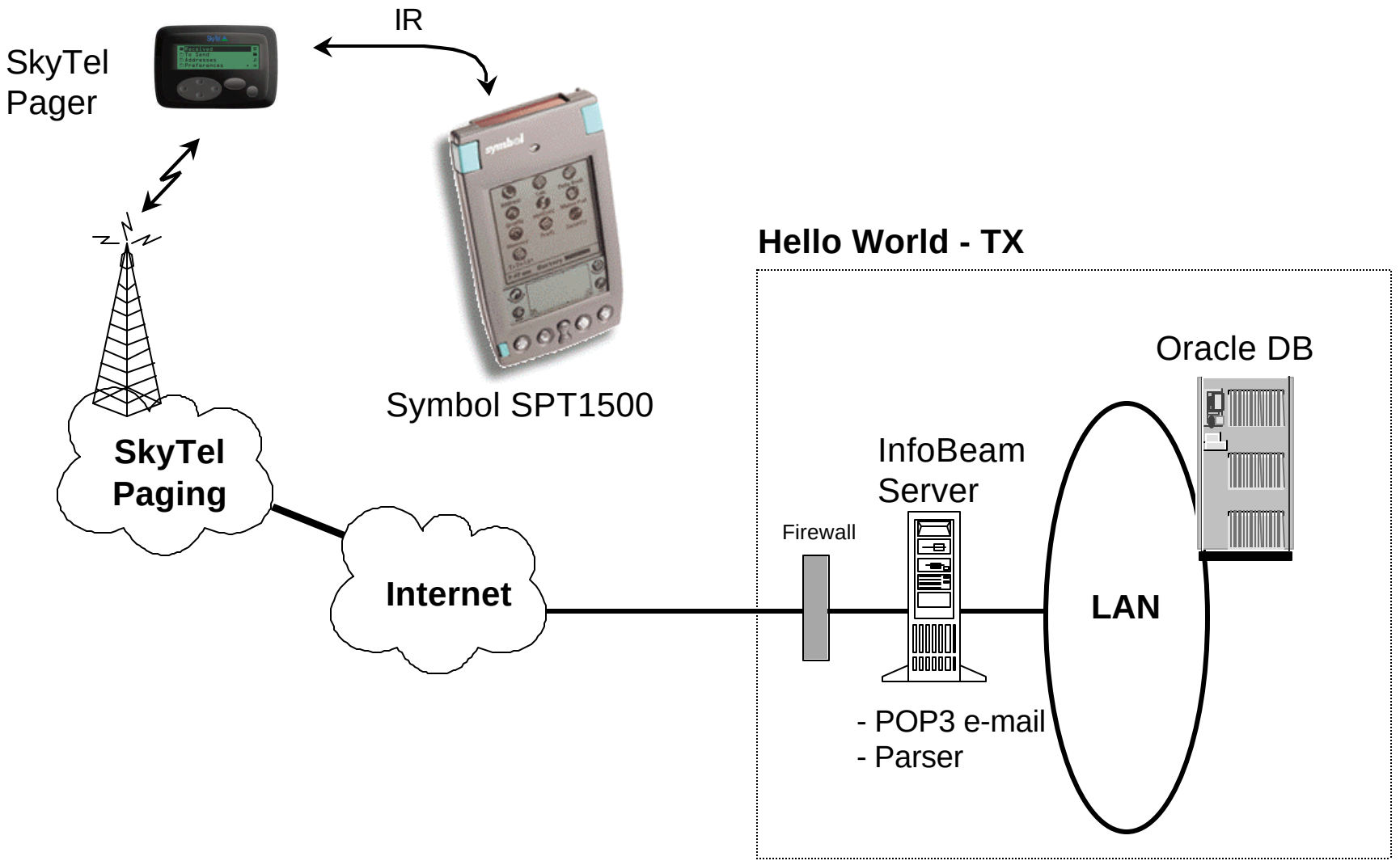
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Case Study #1: Hello World

PalmSource 99

- Hello World technicians stock calling card machines with calling cards
  - Problems with inventory shrinkage
- Calling cards are scanned w/ Symbol SPT-1500
  - Accumulated scans beamed to SkyTel AccessLink II pager
  - Sent via SMTP to InfoBeam server
- InfoBeam server parses e-mail, deposits inventory data in Oracle database at Hello World

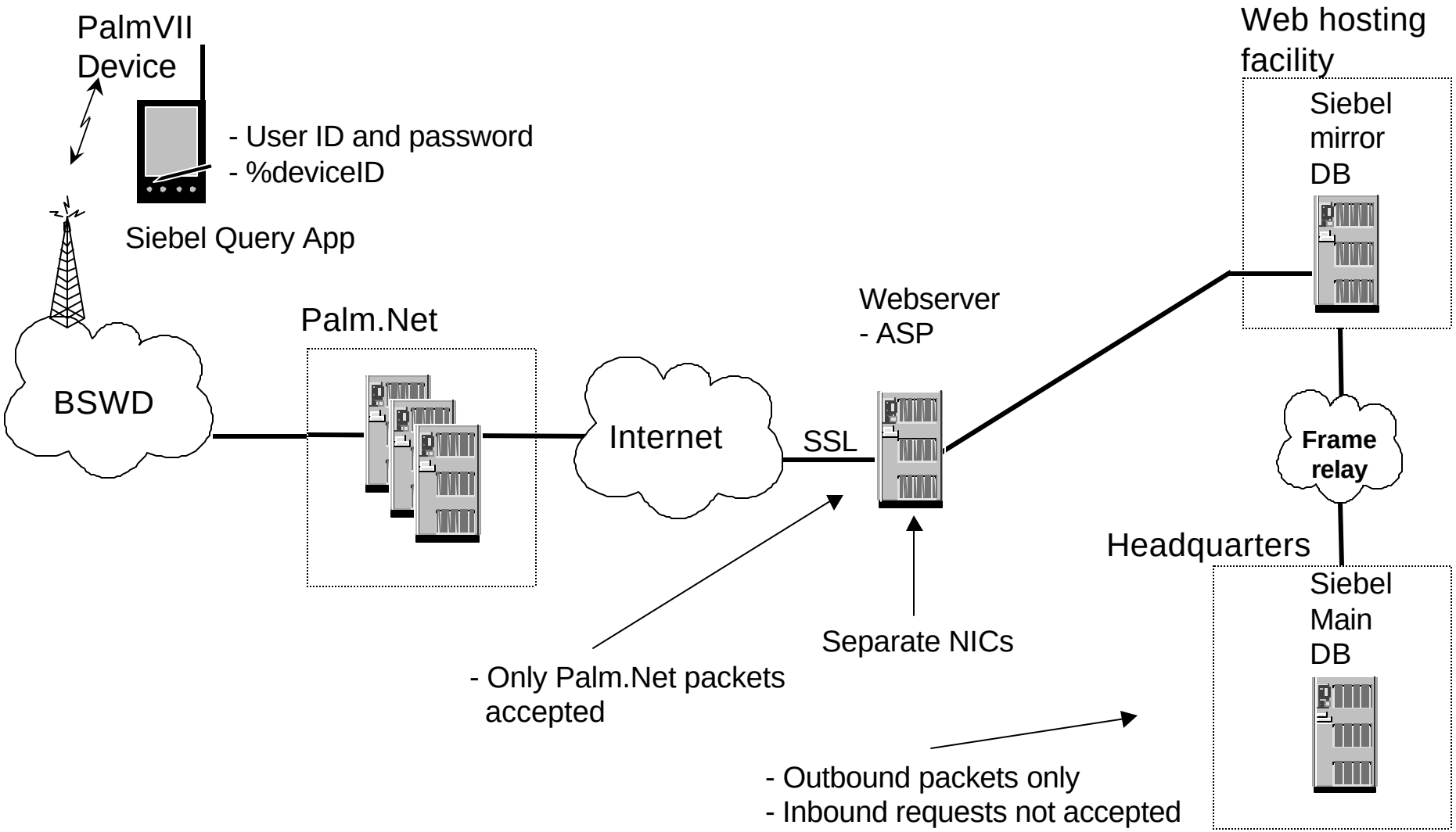
Case Study #1: Hello World



Case Study #2: Frank Russell

- PalmVII access to Siebel SFA database
- Mobile sales force—selling financial services
- PQA-based application
  - Query customer name
  - Query company name
  - Wild card searches
- Currently using a database replication solution; final product may use a secure firewall “pinhole” solution

Case Study #2: Frank Russell





Q&A


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